SA 2 - Statistical Theory

Question 2

1. given points: (3, -2) and (-1, 6)

a) $m = (y_2 - y_1) / (x_2 - x_1)$

$m = (6 - (-2)) / (-1 - 3)$

$m = 8 / -4 = \boxed{-2}$

b) using (3, -2)

$y = (mx + b)$

$y - (-2) = -2(x - 3)$

$y + 2 = -2x + 6$

$\boxed{y = -2x + 4}$

c) y-intercept: $x = 0$

$y = -2(0) + 4$

$y = 4$

(0, 4)

x-intercept: $y = 0$

$0 = -2x + 4$

$2x = 4$ (divide 2 both sides)

$x = 2$

(2, 0)

d) substitute the (3, -2):

$-2 = -2(3) + 4$

$-2 = -6 + 4$

$-2 = -2$ *(verified: correct)*

(-1, 6):

$6 = -2(-1) + 4$

$6 = 2 + 4$

$6 = 6$ *(verified: correct)*

2. a) $\Sigma X = 3 + 5 + 6 + 8 + 9 + 11 = 42$

$\Sigma Y = 2 + 3 + 4 + 6 + 5 + 8 = 28$

$\Sigma X^2 = 3^2 + 5^2 + 6^2 + 8^2 + 9^2 + 11^2$

$\Sigma X^2 = 9 + 25 + 36 + 64 + 81 + 121 = 336$

$\Sigma XY = (3x2) + (5x3) + (6x4) + (8x6) + (9x5) + (11x8)$

$\Sigma XY = 6 + 15 + 24 + 48 + 45 + 88 = 226$

b) $Y = a + bX$, $n = 6$

$b = [n\Sigma XY - \Sigma X\Sigma Y] / [n\Sigma X^2 - (\Sigma X)^2]$ $\quad$ $a = [\Sigma Y - b\Sigma X] / n$

$b = [6(226) - (42)(28)] / [6(336) - (42)^2]$ $\quad$ $a = [28 - (0.7143)(42)] / 6$

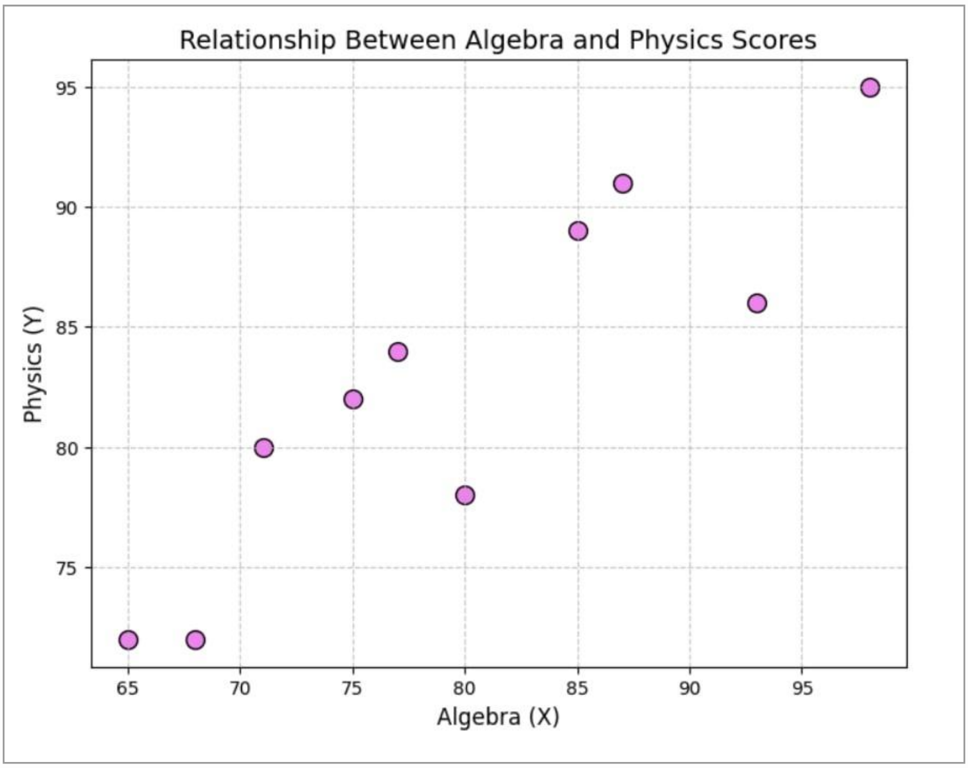
$b = [1356 - 1176] / [2016 - 1764]$ $\quad$ $a = (28 - 30) / 6$

$b = 180 / 252$ $\quad$ $a = -0.333$

$b = 0.7143$ $\quad$ $\boxed{\hat{Y} = 0.71X - 0.33}$

3. $n = 10$, $\Sigma X = 800$, $\Sigma X^2 = 64962$, $\Sigma XY = 66149$

a)



b) $b = [10(66149) - (800)(819)] / [10(64962) - (800)^2]$ $\quad$ $a = [819 - (0.6538)(800)] / 10$

$b = [661490 - 655200] / [649620 - 640000]$ $\quad$ $a = [819 - 523.04] / 10$

$b = 6290 / 9620$ $\quad$ $a = 29.596$

$b = 0.6538$

$\boxed{\hat{Y} = 0.65X + 29.60}$

c) predict physics grade if $X = 75$

$\hat{Y} = 0.65(75) + 29.60$

$\hat{Y} = 48.75 + 29.60$

$\hat{Y} = 78.35$

d) predict algebra grade if $Y = 95$

$\Sigma Y^2 = 67603$

$b_{xy} = 6290 / [10(67603) - (819)^2] = 6290 / 5269 = 1.1938$

$a_{xy} = [800 - (1.1938)(819)] / 10 = -17.77$

$\check{X} = 1.19(95) - 17.77 = 95.28$

e) The relationship between strength and direction has strong and positive.
The higher grades that can get for algebra is generally correspond to higher grades also in physics.